UNIVERSIDAD TECNOLÓGICA EL RETOÑO

Major:
Information Technologies and Digital Innovation Engineering

Term:
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Subject:
Disruptive Technologies

Group:
LITD9A

Activity 4. Digital Transformation Blueprint

Student's ID and name:

Teacher:
Dr. Sergio Valadez Godínez

Date: mm/dd/yyyy

1. Activity Type
Team Work

2. Activity Context and Continuity
In previous activities, teams identified a real-world challenge, explored emerging technologies, analyzed scientific and technological evidence, and evaluated technology adoption potential through TRL, feasibility, viability, impact, and adoption assessments.

The project will now move from:
Technology Selection → Digital Transformation Design

This activity will not repeat:

- Problem discovery.
- Stakeholder identification.
- Scientific intelligence.
- Technology scouting.
- Innovation observatory activities.
- TRL assessment.
- Technology adoption assessment.

The selected technology or technology combination identified during **Technology Adoption Assessment** will become the input for this activity.

The objective is no longer determining **what technology should be used**, but rather:
How must people, processes, technology, organizational culture, and change management evolve in order to successfully adopt the proposed solution?

Students will develop a Digital Transformation Blueprint that serves as a strategic roadmap for implementation.

This activity applies equally to:

- Data Science projects.
- Software systems.
- Smart city initiatives.
- Health technologies.
- Sustainability solutions.
- Computer Vision systems.
- Digital Image Processing projects.
- AI-based identification or pre-diagnostic systems.

Students are not expected to build prototypes during this activity.

3. Objective

To design a Digital Transformation Blueprint that aligns people, processes, technology, organizational culture, and change management strategies in support of the selected technological solution.

Additionally, students will document the transformation strategy using GitHub and collaborative project documentation practices to ensure traceability, reproducibility, and professional project organization.

4. Expected Learning Outcomes

Knowledge

Students will be able to:

- Explain the concept of Digital Transformation.
- Differentiate digitization, digitalization, and digital transformation.
- Explain the role of people, processes, and technology.
- Explain organizational culture and cultural transformation.
- Explain resistance to organizational change.
- Describe change management approaches.
- Explain technology adoption within organizations.
- Recognize barriers to transformation.
- Explain the relationship between innovation and organizational capabilities.

Skills

Students will be able to:

- Analyze organizational readiness.
- Identify stakeholders affected by transformation.
- Map business processes.
- Assess digital transformation opportunities.
- Identify capability gaps.
- Develop transformation roadmaps.
- Design change management strategies.
- Create adoption plans.
- Communicate transformation proposals.
- Document organizational and technological strategies.

Professional Attitudes

Students will develop:

- Strategic thinking.
- Systems thinking.
- Innovation mindset.
- Leadership awareness.

- Change readiness.
- Organizational awareness.
- Stakeholder empathy.
- Evidence-based decision making.

5. General Instructions

- Continue working with the same project developed throughout the semester.
- Use the technology selected during Technology Adoption Assessment.
- Do not perform a new technology scouting process.
- Do not reassess technological maturity.
- Focus on organizational transformation rather than technology evaluation.
- Consider the impact on people, processes, technology, and culture.
- Support conclusions with evidence generated in previous activities.
- Continue using the GitHub repository created for the project.
- Document findings using Markdown files or reports stored in the repository.
- Use diagrams whenever appropriate.
- Consider adoption risks and resistance factors.
- Distinguish current organizational conditions from desired future conditions.

Recommended Project Environment

Core Tools

- GitHub Repository
- Notion (recommended)
- Jupyter Notebook (if analytical evidence is required)
- Draw.io / diagrams.net
- Miro (optional)
- PowerPoint / Canva (optional)

Recommended Project Structure

```
project/
│
├── docs/
│   ├── transformation/
│   │   ├── digital_transformation_blueprint.md
│   │   ├── process_map.png
│   │   ├── stakeholder_analysis.pdf
│   │   └── roadmap.png
│
├── data/
│
├── notebooks/
│
├── reports/
│
├── README.md
│
└── requirements.txt
```

PHASE 1: CURRENT STATE ASSESSMENT

Step 1 – Organizational Context Analysis

Describe the current context in which the proposed solution will operate.

Analyze:

People

- Users.
- Employees.
- Stakeholders.
- Decision makers.

Processes

- Current workflows.
- Existing procedures.
- Inefficiencies.

Technology

- Existing systems.
- Current tools.
- Technical limitations.

Culture

- Organizational attitudes.
- Innovation climate.
- Readiness for change.

Expected Output

Current State Assessment Matrix

Dimension	Current Situation	Key Challenges
People		
Processes		
Technology		
Culture		

Step 2 – Transformation Opportunity Analysis

Based on the selected solution, identify opportunities for transformation.

Examples:

- Process automation.
- Decision support.
- Improved efficiency.

- Better user experiences.
- Faster service delivery.
- Improved diagnostics.
- Better information access.

For Computer Vision or Image Processing Projects
Examples may include:

- Automated image interpretation.
- Decision support for specialists.
- Faster screening processes.
- Reduction of repetitive human tasks.
- Improved identification accuracy.

Expected Output
Transformation Opportunity Matrix

Opportunity	Expected Benefit	Affected Area

PHASE 2: DIGITAL TRANSFORMATION DESIGN

Step 3 – Future-State Blueprint
Design the future scenario after implementing the proposed solution.

Describe:
Future People

- New responsibilities.
- New competencies.
- Training requirements.

Future Processes

- Improved workflows.
- Automation opportunities.
- Decision-making improvements.

Future Technology

- New systems.
- New capabilities.
- Data flows.

Future Culture

- Desired behaviors.
- Innovation mindset.
- Digital adoption.

Expected Output
Future-State Blueprint
Include a diagram illustrating:

```
People
   ↓
Processes
   ↓
Technology
   ↓
Value Creation
```

Step 4 – Digital Capability Gap Analysis
Compare the current state with the future state.

Identify:

- Skills gaps.
- Process gaps.
- Technology gaps.
- Cultural gaps.

Expected Output
Capability Gap Matrix

Area	Current State	Desired State	Gap

PHASE 3: CHANGE MANAGEMENT

Step 5 – Stakeholder Adoption Strategy
Identify how stakeholders will be affected.

For each key stakeholder group:

- Expected benefits.
- Possible concerns.
- Resistance factors.
- Adoption strategies.

Expected Output
Stakeholder Adoption Matrix

Stakeholder	Expected Benefit	Possible Resistance	Adoption Strategy

Step 6 – Change Management Plan
Design a practical strategy to facilitate adoption.

Include:
Communication
How will the change be communicated?

Training
What new competencies are required?

Support
How will users receive assistance?

Monitoring
How will adoption progress be evaluated?

Expected Output
Change Management Plan

PHASE 4: TRANSFORMATION ROADMAP

Step 7 – Digital Transformation Roadmap
Develop an implementation roadmap.

Suggested stages:
Stage 1
Preparation

Stage 2
Capability Development

Stage 3
Pilot Implementation

Stage 4
Deployment

Stage 5
Continuous Improvement

For each stage, identify:

- Objectives.
- Key activities.
- Responsible stakeholders.
- Expected outcomes.

Expected Output
Digital Transformation Roadmap
The roadmap should include a timeline or phased implementation diagram.

Step 8 – Transformation Assessment
Evaluate the proposed transformation.

Assess:
Strategic Alignment
Does the transformation support the project goals?

Organizational Readiness
Is the organization prepared?

Expected Value
What value is expected?

Main Risks
What could prevent success?

Sustainability
Can the transformation be maintained over time?

Finally classify the project as:

READY FOR TRANSFORMATION
Strong alignment and clear roadmap.

CONDITIONALLY READY

Important gaps remain.

ADDITIONAL PREPARATION REQUIRED
Significant readiness problems remain.

Expected Output
Digital Transformation Assessment and Strategic Recommendation

GitHub and Documentation Requirements
The repository must contain:

```
project/
│
├── docs/
│   └── transformation/
│
├── reports/
│
├── README.md
│
└── roadmap/
```

Commit Examples

Add current state assessment
Create transformation opportunity analysis
Develop future-state blueprint
Add stakeholder adoption strategy
Document transformation roadmap

Avoid:

changes
final
last version

6. Deliverables

- Current State Assessment Matrix
- Transformation Opportunity Matrix
- Future-State Blueprint
- Capability Gap Matrix
- Stakeholder Adoption Matrix
- Change Management Plan
- Digital Transformation Roadmap
- Digital Transformation Assessment and Strategic Recommendation
- GitHub Repository
- Updated README.md

- Git Commit History

7. Evaluation Criteria

Criterion	Weight
Current State Assessment	10%
Transformation Opportunity Analysis	15%
Future-State Blueprint	20%
Capability Gap Analysis	10%
Stakeholder Adoption Strategy	15%
Change Management Plan	10%
Transformation Roadmap	15%
Documentation, GitHub, and Traceability	5%
Total	**100%**

8. References

Use **APA 7th Edition** format.

Include:

- Digital Transformation references.
- Change Management references.
- Organizational Culture references.
- Technology Adoption references.
- Industry and government reports.
- Scientific articles supporting the transformation approach.

1. GitHub Repository and Project Context
Instructions: Describe the repository organization, project context, selected technology, and purpose of the transformation initiative.
2. Current State Assessment
Instructions: Describe people, processes, technology, and culture in the current situation.
3. Transformation Opportunities
Instructions: Explain the opportunities enabled by the proposed technology and how they create value.

4. Future-State Blueprint

Instructions: Present the desired future state and explain the interaction among people, processes, technology, and culture.

5. Capability Gap Analysis

Instructions: Identify what capabilities must be developed to reach the desired future state.

6. Stakeholder Adoption Strategy

Instructions: Analyze benefits, concerns, resistance factors, and adoption strategies for stakeholders.

7. Change Management Plan

Instructions: Describe communication, training, support, and monitoring strategies.

8. Digital Transformation Roadmap

Instructions: Present the phased transformation plan and expected outcomes.

9. Transformation Assessment

Instructions: Evaluate readiness, value, risks, sustainability, and strategic fit.

10. Conclusion

Instructions: Summarize the transformation strategy, expected impact, major risks, and readiness level.

11. References

Instructions: Include all sources using APA 7th Edition.